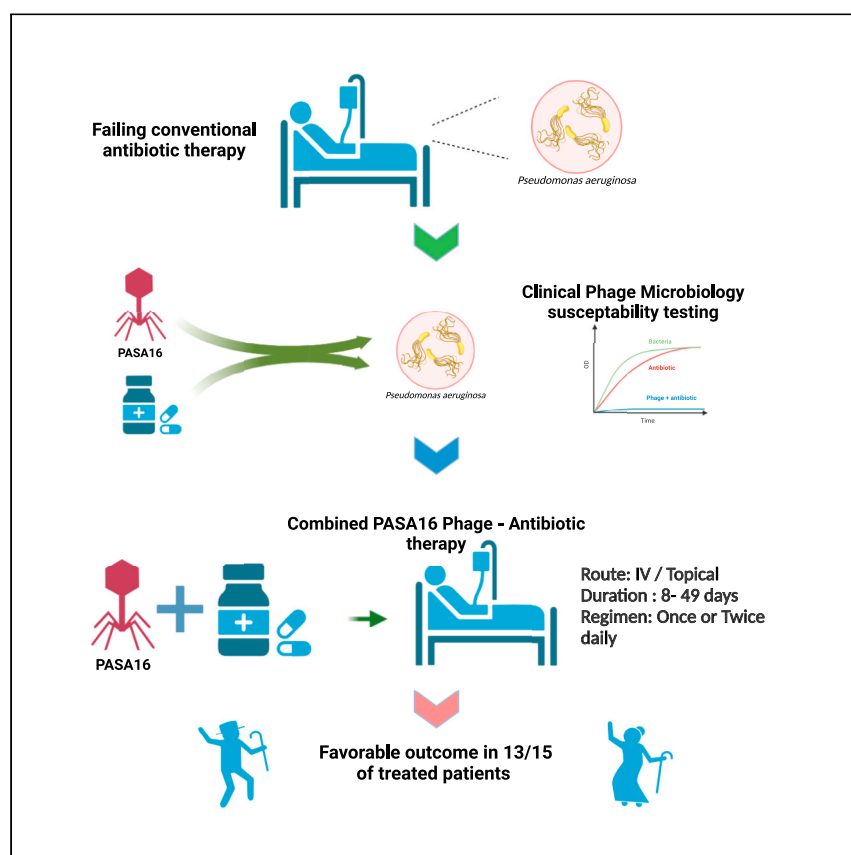


Clinical Advances

Refractory *Pseudomonas aeruginosa* infections treated with phage PASA16: A compassionate use case series



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Highlights

A global *Pseudomonas aeruginosa* infection series that failed conventional treatment

All were compassionate use treated using a single anti-pseudomonal phage, PASA16

For all treatments, *Pseudomonas* phage susceptibility was demonstrated in the lab

PASA16 phage therapy demonstrated promising outcomes with minimal side effects

Resistant and persistent infections pose a major challenge in modern medicine. Onallah et al. report the largest case series of PASA16 phage compassionate use for the treatment of refractory severe *P. aeruginosa* infections. Study findings suggest that the use of PASA16 phage and antibiotics may lead to favorable outcomes.



Translation to Patients

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Refractory *Pseudomonas aeruginosa* infections treated with phage PASA16: A compassionate use case series

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SUMMARY

Background: A growing number of compassionate phage therapy cases were reported in the last decade, with a limited number of clinical trials conducted and few unsuccessful clinical trials reported. There is only a little evidence on the role of phages in refractory infections. Our objective here was to present the largest compassionate-use single-organism/phage case series in 16 patients with non-resolving *Pseudomonas aeruginosa* infections.

Methods: We summarized clinical phage microbiology susceptibility data, administration protocol, clinical data, and outcomes of all cases treated with PASA16 phage. In all intravenous phage administrations, PASA16 phage was manufactured and provided pro bono by Adaptive Phage Therapeutics. PASA16 was administered intravenously, locally to infection site, or by topical use to 16 patients, with data available for 15 patients, mainly with osteoarticular and foreign-device-associated infections.

Findings: A few minor side effects were noted, including elevated liver function enzymes and a transient reduction in white blood cell count. Good clinical outcome was documented in 13 out of 15 patients (86.6%). Two clinical failures were reported. The minimum therapy duration was 8 days with a once- to twice-daily regimen.

Conclusions: PASA16 with antibiotics was found to be relatively successful in patients for whom traditional treatment approaches have failed previously. Such pre-phase-1 cohorts can outline potential clinical protocols and facilitate the design of future trials.

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CONTEXT AND SIGNIFICANCE

Phage therapy has re-emerged in the last few years as a potential solution for antimicrobial-resistant and non-resolving infections. While phages are being used on compassionate use bases, clinical trials for phage therapy are still lacking. This study reports the largest case series of consecutive patients with severe *Pseudomonas aeruginosa* infections who received compassionate use treatment using a single specific phage, PASA16. All 16 patients were treated after demonstrating that their infective agent was susceptible to both the phage alone and in combination with antibiotics. The study findings demonstrate a favorable outcome in >80% of treated patients, with minimal side effects. Based on these findings, further compassionate use cases and clinical trial protocols could follow.

INTRODUCTION

In recent years, bacteriophage (phage) therapy has grown more common as a salvage treatment for non-resolving infections,^{1,2} and the number of reports describing phage-based treatments is increasing in the medical literature.^{3–5} Phages are viruses that attack bacteria without current known harm to human cells or the microbiome and are mainly well tolerated among patients.³ As such, they can be used as a potential adjunct to antibiotics, particularly when the latter prove to be ineffective.^{6,7} Several case series have been published on the use of phage therapy to treat respiratory, mycobacterial, and osteoarticular infections.^{8–15} Treatment of infected immunocompromised patients has also been promising.^{7,16}

Access and use of phages for therapy vary across countries. In Poland, Georgia, and Russia phages are approved and have been employed for decades.¹⁷ In Belgium, pharmacies are allowed to make phage preparations.^{18,19} In other countries, such as the United States, France, or Australia, phages have been administered on a compassionate basis with permission from local authorities. More recently, clinical trials have been initiated,^{20–22} and trials to support a wide range of applications of phages are still lacking.^{23,24} However, because of their replicative nature, complex pharmacodynamics and pharmacokinetics, and complex interactions with their bacterial host and the immune system, phase 0 or phase 1 clinical trials are needed to supplement the information in papers that have been published to date, which may suffer from reporting bias. With that in mind, as expected, most of the cases reported until now had favorable outcomes, which may reflect reporting biases and not necessarily a true favorable outcome.^{25–27} To address this potential bias, novel protocols such as the PHAGEFORCE and STAMP studies are currently being implemented with the aim of reducing such reporting biases.^{20,28} Here, we report a cohort study on the use of a single phage (PASA16) in 16 consecutive patients from around the world for the treatment of refractory *Pseudomonas aeruginosa* infections.

RESULTS

The first patient treated with PASA16

In 2019, a 46-year-old male patient was referred to the Infectious Diseases clinic at Hadassah-Hebrew University Medical Center for possible phage therapy for a chronic non-resolving bone infection. The patient had multiple traumatic injuries 25 years earlier. During his prior hospital admissions, he developed osteomyelitis in his distal left tibia with multiple isolation of *P. aeruginosa*. In recent years, he experienced few exacerbations per year accompanied by pain, swelling, and purulent drainage. Imaging studies suggested that over time his infection had extended both upward in his tibia and downward to his foot. Infectious disease physicians and orthopedic surgeons who reviewed the patient's case recommended amputation of his left leg, but he refused to have this done and sought alternative treatments. In response to this, phages at the Israeli Phage Bank were screened. PASA16 anti-*P. aeruginosa* lytic phage was found to kill the bacteria isolated and to synergize with antibiotics. Bacteriophage PASA16 was sent to Adaptive Phage Therapeutics (APT) for manufacture so that it could be administered intravenously. Treatment was initiated in July 2019 and the patient received twice-daily (BID) intravenous (i.v.) phage therapy for 2 weeks with ceftazidime three times a day. Four years have passed since the patient received treatment. During the treatment he reported dizziness and tiredness during the administration of i.v. PASA16. He remains in remission ever since and has had no exacerbations (case 1, [Table 4](#) and case 1, [supplemental information](#)).

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Continued

Table 1. Bacterial isolate clinical microbiology performed at IPTC

Case #	Center performing clinical susceptibility testing	Plaque assay	Phage kinetics planktonic (h)	Antibiotics used for checkerboard	Combination with antibiotics/phages kinetics Kinetics planktonic (h)
1	IPTC	clear plaques	regrow (at 14 h)	ceftazidime	no regrow (at 20 h)
3	IPTC	turbid plaques	regrow (at 10 h)	meropenem gentamicin ciprofloxacin	no regrow (at 10 h)
4	IPTC	clear plaques	regrow (at 14 h)	colistin piperacillin/ tazobactam piperacillin	regrow (at 14 h)
5	IPTC	clear plaques	regrow (at 35 h)	ceftazidime	no regrow (at 45 h)
7 strain 1	IPTC	clear plaques	regrow (at 12 h)	meropenem piperacillin	no regrow (at 25 h) regrow (at 18 h)
strain 2			regrow (at 3 h)	meropenem piperacillin	regrow (at 3 h) regrow (at 3 h)
strain 3			regrow (at 3 h)	meropenem piperacillin	no regrow (at 25 h) no regrow (at 25 h)
strain 4			regrow (at 8 h)	meropenem piperacillin	regrow (at 15 h) regrow (at 8 h)
strain 5			regrow (at 12 h)	meropenem piperacillin	no regrow (at 25 h) no regrow (at 25 h)
strain 6			regrow (at 12 h)	meropenem piperacillin	no regrow (at 25 h) no regrow (at 25 h)
8	IPTC	clear plaques	regrow (at 2 h)	gentamicin ciprofloxacin ceftazidime	regrow (at 22 h)

Detailed phage microbiology was performed in three locations. Each location has used different methodology, and thus each is reported in a separate table. This table presents detailed phage clinical microbiology results for each case performed at the IPTC. Results present both plaque assay and kinetic assay. In liquid growth kinetics, the kinetics of phage-bacteria and the combination of phage-antibiotics with bacteria were tested. Bacterial regrowth was observed in the phage-bacteria kinetics, while no growth was seen in the phage-antibiotics combination.

Clinical phage microbiology of cases and PASA16

Testing bacterial susceptibility to an antimicrobial is the foundation for a successful therapy; however, no standardization yet exists for bacterial susceptibility to phages, and as yet there is no single method that is known to have a good prediction of clinical outcome. For cases described here, phage susceptibility was carried out at the Israeli Phage Therapy Center (IPTC) (Table 1), at APT (Table 2), or at the Iredell lab at the Westmead Institute for Medical Research (WIMR) (Table 3). At the IPTC, the process for all patients treated with PASA16 began with a plaque assay to identify phages with clear plaques and high titers against the bacterial isolate. Thereafter, a kinetic assay was conducted in liquid culture to assess phage-bacterial interactions and time to bacterial regrowth, together with potential synergistic/additive or antagonistic effects with antibiotics. Only patients with susceptibility of bacteria in all methods were selected for treatment. At the WIMR, the first step was to conduct a plaque assay and calculate the efficiency of plating. Thereafter, phage kinetics were tested to observe bacterial regrowth. At APT, experiments were conducted to study the kinetics of phage-bacterial and phage-antibiotic interactions in both planktonic and biofilm conditions. The regrowth of bacteria was also evaluated in both conditions. In most cases treated with PASA16, when the kinetic assay was performed, it exhibited no regrow in the presence of phage and antibiotics for up to 48 h or late regrowth (>8 h) of bacteria after lysis (Table 1).

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Table 2. Bacterial isolate clinical microbiology performed at APT

Case #	Center performing clinical susceptibility testing	Phage kinetics		Antibiotics used for checkerboard	Combination with antibiotics/phages kinetics	
		Kinetics planktonic (h)	Kinetics biofilm (h)		Kinetics planktonic (h)	Kinetics biofilm (h)
2	APT	6	–	–	–	–
6	APT	data are missing				
9	APT	data are missing				
10	strain 1 APT	46	14	cefepime	46	19
				PaWRA02Phi83	46	45
		46	4	cefepime	46	38
				PaWRA02Phi83	46	43
		45	15	cefepime	45	15
				PaWRA02Phi83	45	43
		46	34	cefepime	46	17
				PaWRA02Phi83	10	15
11	strain 1 APT	12	1	gentamicin	45	31
				imipenem	45	31
	strain 2	13	8	gentamicin	45	27
				imipenem	45	27
12	APT	35	9	ceftazidime	35	12
13	APT	42	1	piperacillin/tazobactam	42	34
				meropenem	42	12
				ampicillin	42	21
14	APT	43	13	ceftazidime	–	10
				gentamicin	–	13

Detailed phage clinical microbiology results for each case performed at APT. Results present phage kinetic assay in planktonic and biofilm conditions. In liquid growth kinetics, bacterial regrowth was much faster than in biofilm. Similar results were observed in most cases in kinetic assay of phage-antibiotic combination.

PASA16 therapy cases in Israel and abroad

Since the initial patient was treated in 2019, phage PASA16 was given on a compassionate basis to a total of 16 patients who ranged in age from 7 to 96 years (Figure S1). Brief overviews of each treatment, including the patients' characteristics, are presented in Table 4 (detailed narratives of each case are presented in supplemental information, in Table S2, and in case descriptions). Five patients in Israel were treated with PASA16, while the remaining patients received treatment abroad, all with the assistance of APT in preparing a suitable formulation. As part of this study, two patients underwent phage therapy according to the STAMP protocol. Separate publications will be made available upon trial completion regarding the pharmacokinetic and pharmacodynamic parameters as well as the immunological responses observed in these patients during the course of phage therapy. *P. aeruginosa* infections were detected in all 15 patients. Over half of the patients who received i.v. PASA16 treatments suffered from bone, joint, and prosthetic device infections (i.e., osteomyelitis). The remaining patients suffered from biofilm-associated bacteremia, post-liver-transplant infections, skin and soft tissue infections, otitis media, and sinusitis. Nine out of the 15 patients had *P. aeruginosa* antibiotic-resistant infections.

Treatment and follow-up

The dosage of phage, route(s) of administration, and duration of treatment varied from case to case (Table 4) and were determined based on the infection site and

Table 3. Bacterial isolate clinical microbiology performed at Iredell lab, Westmead Institute for Medical Research

Case #		Center performing clinical susceptibility testing	Plaque assay	Efficiency of plating (%)	Phage kinetics planktonic (h)
15	strain 1	Iredell lab, WIMR	clear plaques	22	regrow (at 7 h)
	strain 2			13	regrow (at 7 h)
	strain 3			13	regrow (at 7 h)
	strain 4			10	regrow (at 7 h)
16	strain 1	Iredell lab, WIMR	clear plaques	108	regrow (at 7 h)
	strain 2			75	regrow (at 7 h)
	strain 3			162	regrow (at 18 h)

Detailed phage clinical microbiology results for each case performed at Iredell lab, Westmead Institute for Medical Research. Results present spot test assay, efficiency of plating, and phage kinetic assay.

the treating clinician's recommendations, due to the lack of firm recommendations. Treatment duration ranged widely, from 8 days to 49 days. Phages were administered once daily (QD) ($n = 4$), twice daily (BID) ($n = 8$), or as a combination of the two regimens ($n = 3$). For intravenous administration, two batches of PASA16 were produced, and dosing was determined based on the endotoxin level of each batch. In four cases, intraoperative or intralesional injections were given prior to the initiation of i.v. treatment, and in two cases, a topical administration was applied because of driveline infections and sinusitis infections. Clinicians determined the optimal antibiotic for treatment based on the results of the phage-antibiotic kinetic assay. In two cases (cases 3 and 7), antibiotics were administered despite the infection being determined to be resistant based on the antibiogram results, but they were found to have a synergistic effect with phage on the clinical phage microbiology assay. Seven of the cases were treated with a combination of antibiotics that were shown to be effective and possess synergism with PASA16 according to the clinical phage microbiology assay results while, for the remainder, combination with antibiotics was empirical. Since the compassionate treatments spanned many years and were uniquely tailored for each patient, the follow-up periods varied widely from <1 month to 5 years depending on patients' availability for follow-up. One case was lost to follow-up whereby no further data were collected.

Safety and outcomes

Clinical recovery appeared to take place in six cases, and remission was seen in an additional seven patients, most of whom had infections of osteomyelitis and prosthetic devices. Unfortunately, two patients out of the 15 died (cases 2 and 3); both had severe underlying conditions and drug-resistant polymicrobial infections. Based on the data presented, there was an 86.6% cure/remission rate (13 out of 15). Microbiological recovery was identified in seven cases out of the 15. There was one complete failure and seven cases with a "not specified" assessment because no cultures were taken (Table 4), suggesting at least a 46.6% microbiological response rate (7 out of 15). During the entire treatment period, physicians and hospital staff closely monitored patients for adverse events and safety signals. PASA16 was generally well tolerated by patients. Three patients experienced a reversible elevation in liver enzymes up to twice the normal upper limit ($2 \times \text{ULN}$). Furthermore, two patients had a minor decrease in white blood cell count (WBC) (case 2: reduction from $6.7 \times 10^9/\text{L}$ to $5.3 \times 10^9/\text{L}$, and case 4: reduction from $6.3 \times 10^9/\text{L}$ to $3.0 \times 10^9/\text{L}$) (normal range: $4.5\text{--}11.0 \times 10^9/\text{L}$). One patient had a gradual decline in hemoglobin as well (case 4: reduction from 124 mg/L at baseline to 105 mg/L on day 11 of treatment) (normal range for women: 121–151 mg/L), and one reported dizziness and tiredness. We

Table 4. Overview of the 16 cases of PASA16 bacteriophage therapy

Case #	Clinical indication	Age/ gender	Source of bacterial isolation	Antibiotic susceptibility	Route of administration and duration	Regimen of phage therapy	Phage dose (PFU/mL)	Combination with standard-of-care antibiotics (name and duration)	Combination with other phages	Adverse events	Outcome of patient	Microbiological outcome of infection	Follow-up time	Published case
1	osteomyelitis	46/M	wound tibial infection	MDR	i.v., 14 days	BID	1.72×10^{11}	ceftazidime	none	dizziness and tiredness	remission	not specified	4 years	
2	post-liver-transplant infection and sepsis	39/F	abdominal mesh	XDR	i.v., 9 days	QD	1.72×10^{11}	polymyxin B, eravacycline, micafungin	<i>Klebsiella pneumoniae</i> phage (KpKH52Phi07B)	reduced WBC count, increase in pressors requirement	deceased	not specified	lost to follow-up	
3	bacteremia	10/F	prosthetic device infection (Berlin heart EXCOR)	MDR	i.v., 49 days	BID	1.72×10^{11}	meropenem	none	elevated LFT: 2x ULN, high fever	deceased	failure	2 months	
4	osteomyelitis	7/F	surgical debridement	XDR	i.v., 14 days	QD (days 1, 2 and 4–7), BID (days 3 and 8–14)	1.34×10^{11}	colistin, aztreonam	none	elevated LFT: 2x ULN, reduced WBC and Hb	remission	not specified	8 months	Khatami et al. ²⁵
5	mastoiditis	69/M	ear infection	S	i.v., 14 days	BID	1.72×10^{11}	ceftazidime	none	none	remission	not specified	2 months	
6	joint infection	62/F	synovial fluid	S	intraoperative + i.v., 8 days	BID (3 days), QD (5 days)	1.7×10^9	cefepime, switched later to oral ciprofloxacin	none	none	recovery	recovery	1 month	
7	skin and soft tissue infection	41/F	massive silicon soft tissue infection in back buttocks and thigh	PDR	i.v., 2 courses of 22 and 30 days + topical wound dressing for 30 days	BID, and wound dressing once a day	3×10^{10}	meropenem	none	none	recovery	recovery	17 months	
8	osteomyelitis	32/M	surgery wound	MDR	intraoperative + i.v., 28 days	BID	4.3×10^{10}	ceftazidime/ avibactam, gentamicin	none	none	remission	not specified	6 months	
9	osteomyelitis	25/M	debrided cranial bone and scalp tissue	XDR	i.v., 30 days	BID	1.72×10^{11}	cefiderocol	none	none	recovery	recovery	1 year	Simner et al. ²⁶
10	prosthetic joint infection	96/M	synovial fluid tissue sonicate	S	IA and i.v., 10 days	5 days BID x 2 cycles	3.0×10^9	piperacillin/ tazobactam, cefepime	PaWRA02Phi83 both intraoperative and i.v., 3.0×10^9 PFU	none	remission	not specified	1 year	
11	otitis media	46/F	middle ear	XDR	intra-auricular + i.v., 10 days	QD	2.35×10^{10}	none	PaWRA01Phi11, 1.4×10^9 PFU	none	recovery	recovery	1 month	
12	no information provided													
13	prosthetic device infection (LVAD)	29/M	drainage from the driveline site	MDR	i.v. + topically, 2 weeks	i.v. QD, topically QD	4.7×10^{10}	first week: piperacillin/ tazobactam, vancomycin; second week: piperacillin/ tazobactam, daptomycin; four additional weeks: post-phage therapy with ceftolozane/ tazobactam	SaWQ0488Phi, 2.4×10^8 PFU/mL + ALK PaWRA02Phi87, 3.4×10^9 PFU/mL	elevated ALK	recovery	recovery	3 months	

(Continued on next page)

Table 4. Continued

Case #	Clinical indication	Age/gender	Source of bacterial isolation	Antibiotic susceptibility	Route of administration and duration	Regimen of phage therapy	Phage dose (PFU/mL)	Combination with standard-of-care antibiotics (name and duration)	Combination with other phages	Adverse events	Outcome of patient	Microbiological outcome of infection	Follow-up time	Published case
14	chronic osteomyelitis of the pelvis	43/F	drainage of abscess	MDR	intraoperative + i.v., 6 weeks	QD, 5 times a week; single bone injection; single peri-acetabular soft tissue injection	peri-acetabular soft tissue injection, 7.23×10^{10} ; bone injection, 2.17×10^{10} ; i.v. 1 mL of 4.7×10^{10}	meropenem	none	none	remission	recovery	7 months	
15	sinusitis	16/M	blood and maxillary sinus tissue and pus	S	intranasal wash, 14 days	QD for 2 days, BID for 12 days	1×10^9 nasal wash	ceftolozane/tazobactam, ciprofloxacin, and one dose of tobramycin	Pae7 (i.v.), 1×10^9 PFU/mL	none	recovery	recovery	4 months	
16	sinusitis	57/F	pansinusitis	S	intranasal wash, 14 days	BID for 14 days	3×10^9 nasal wash	oral ciprofloxacin and i.v. piperacillin/tazobactam	Pae7 (i.v.), 1×10^9 PFU/mL + Pae8 (nasal wash), 3×10^9 PFU/mL	none	partial remission	not specified	3 months	

Summary of the 16 cases of PASA16 bacteriophage therapy. PFU, plaque-forming units; M, male; F, female. S, susceptible; QD, once daily; BID, twice daily; MDR, multidrug resistant; XDR, extensively drug resistant; PDR, pan-drug resistant; LFT, liver function test; WBC, white blood cell count; Hb, hemoglobin; ALK, alkaline phosphatase.

cannot correlate those adverse events with the PASA16 administration, and they could be related to the patients' background conditions.

DISCUSSION

Bacteriophage therapy is currently used to treat refractory infections often seen in patients with complex diseases.^{3,29} The need for alternative/adjunctive treatments in such cases is undeniable. Although phage therapy is still used on a compassionate basis, the outcomes of several published bacteriophage therapy cases and the apparent safety of the treatments was encouraging.¹⁷

In this study, we summarize the results of treating 15 out of 16 patients with *P. aeruginosa* infections with PASA16 bacteriophage. While numerous studies have suggested the use of phage cocktails as a potential treatment approach,²² our study focuses on the administration of primarily one phage in combination with standard-of-care antibiotics. However, at this stage it is not yet clear what may be the preferred method of using phages as a single agent or in combination. Each case followed a different protocol that was tailored to the patients' needs. The minimal duration of successful treatment was 8 days and the maximal duration of therapy for a patient was 49 days (case 3). While formal pharmacodynamics and pharmacokinetics may add some useful information, it is not clear when and how such studies will be available for phages and whether findings relevant to one phage will also be adoptable for other phages and infection locations. Based on these data, it appears that a once-daily treatment approach could be effective (as observed in cases 11 and 14) with at least 10 days of therapy, as was suggested by a few researchers.^{2,14,30} However, the optimal dosing scheme for such particular phages is still unclear. In most cases, phages were given along with standard-of-care antibiotics even if these no longer appeared to improve clinical outcomes in the patients or were no longer active *in vitro*. When possible, antibiotics were chosen based on the kinetics assay performed to test for phage-antibiotic synergy (7 cases out of 15). While longer follow-up periods are needed for some cases, recovery or remission were indicated in 86.6% of the cases. This high rate of success is promising and suggests that the phage treatment in general or, specifically, treatment with the phage PASA16 can have significant benefits. However, this may reflect a favorable outcome mainly in prosthetic joint infection and bone and joint infection, since most of the patients described here have such infections. In six cases, a combination of two or more phages was used for treatment of the *P. aeruginosa* infection or for another bacterial pathogen. Interestingly, only four patients had side effects, including moderately elevated liver enzymes and decreases in WBCs and hemoglobin. Therefore, the safety profile of PASA16 based on this cohort was very favorable.

Clinical failure was seen in two patients, who died while they were being treated. The first of these had a transplant infected with *P. aeruginosa*, *Klebsiella pneumoniae*, and *Enterococcus faecium* (case 2, Table 4). The patient received PASA16 phage to treat *P. aeruginosa*. Improvement was noted following the first doses of therapy, but the patient's health declined after administering a *K. pneumoniae* phage 5 days after the initiation of PASA16 phage therapy. The treatment failure may be attributed to the polymicrobial infection and complex phage-phage interactions in such cases.¹¹ APT phage microbiology studies performed after the patient's death showed that the *K. pneumoniae* phage inhibited PASA16; unfortunately, this information was unavailable during the treatment. Thus, this finding may suggest that when approaching treatment using phages in patients with multiple bacterial infections it may be worth performing simulations for possible interactions prior to

treatment. However, the primary factor contributing to the patient's death was multi-organ failure.

The second patient was a 10-year-old girl who developed bacteremia following insertion of a Berlin heart EXCOR (case 3, [Table 4](#)). The patient started phage therapy 3 months after antibiotic treatment of a persistent *P. aeruginosa* blood-borne infection failed. While the patient initially showed improvement, 1 month following the initiation of phage therapy a decline in her general condition was observed that featured high fever, neurological deterioration, and coma. As a result of her worsening neurological state, the patient was declared DNR and passed away shortly after.

In this study, we show the potential of phage therapy in treating persistent infections, with an 86.6% clinical recovery/remission rate. Our results highlight the importance of conducting clinical trials to establish the safety and efficacy of phage therapy. Although based on our results it seems that a relatively short period of treatment may suffice in such cases, there is still a need to define a dosing scheme that will achieve the maximal best outcome. Nevertheless, based on these preliminary results, it appears that many people could benefit from phage treatments.

An additional number of needs must also be met in order for phages to be used optimally, one of which is a standard, predictable, and reliable phage susceptibility scoring and testing system. While many laboratories use similar assays for phage susceptibility testing, the interpretation of results is inconsistent, hindering accurate analysis and selection of phages for use in patients. Standard methods will improve the consistency and accuracy of the choices of phage to be used for treatment.

Limitations of the study are linked to the fact that this is a case series in which there is a major selection bias inherent in the type of patients included in the study. Most infections were of the chronic type, which were stable enough to go through inclusion, supply, and administration of phage. Therefore, conclusions extended to other types of infections may be limited. Additionally, the variability of administration protocols between patients and the lack of pre-treatment structured data collection may limit the ability to draw affirmative conclusions.

In conclusion, it seems that for patients with persistent *P. aeruginosa* infections for whom previous conventional therapy has failed, bacteriophage PASA16 may offer a reasonable and safe treatment option, although larger prospective studies are needed to better define susceptibility, dosing scheme, and duration. Our study offers a potential basis for such studies as well as compassionate use cases in need.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- **RESOURCE AVAILABILITY**
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● **METHOD DETAILS**

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- Phage production for topical use by Iredell Lab at Westmead Institute for medical research
- Phage susceptibility analysis
- Antimicrobial resistance definition
- Outcome definitions

● **QUANTIFICATION AND STATISTICAL ANALYSIS**

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.medj.2023.07.002>.

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AUTHOR CONTRIBUTIONS

Conceptualization, R.N.-P. and R.H.; methodology, H.O., R.N.-P., and R.H.; data collection and analysis, all authors; writing – original draft, H.O., R.H., and R.N.P.; writing – review & editing, all authors; supervision, R.N.P.; H.O., R.H., and R.N.P. had unrestricted access to all data; all authors read and approved the final manuscript, agreed to its submission, and take responsibility for its content.

DECLARATION OF INTERESTS

R.N.P. has been a scientific consultant for BiomX and has participated and served as a PI and on Data Safety Monitoring Boards for a clinical trial by Technophage. N.B. is currently an employee of Pfizer, Israel. M.J.B., J.R.F., B.H., R.H., and S.B. were or are APT (manufacturer of the i.v. PASA16 formulation) employees. K.L.U. is a recipient of NIH grant NIAMS R01 AR082167; he also declares consulting fees from APT not related to the cases described here. R.H. is a recipient of MTEC Foundation Fund MTEC #2019-432. R.H., J.F., and B.H. declare owning stocks or stock options at APT. J.R.F. declares a contract with the United States Defense Health Agency (DHA) to support advancing phage therapy. R.M.L.H. serves on the Takeda advisory board. M.M. was a consultant for Karius, Inc. and Cidara Therapeutics. J.R.I. has been a scientific consultant for Vectura Fertin Pharma. R.C.Y.L., S.L., and A.K. are recipients of the Australia Medical Research Future Fund, Frontiers Stage 1 RFRHPI000017, 2021–2022. R.C.Y.L. and S.L. are recipients of the Australian National Health and Medical Research Council grant Gnt1197534. A.K. is a recipient of the Australian National Health and Medical Research Council grant GNT2008024 and has participated on DSMB for the FluBub Study. R.C.Y.L. had participated on DSMB for Phage Australia as a part of the leadership team, as a chair on the Genomics Scientific Advisory Committee, and as a committee member on the Biobanking Scientific Advisory Committee. S.L. serves as an organizer in Phage Bite, Phage Directory, and as a committee member on Australia Society for Microbiology, Special Interest Group, Phage. P.N.B. received funding from the Royal Australasian College of Physicians for a Cottrell Research Establishment Fellowship. R.B.-A. is a

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INCLUSION AND DIVERSITY

We worked to ensure gender balance in the recruitment of human subjects. We worked to ensure ethnic or other types of diversity in the recruitment of human subjects. One or more of the authors of this paper self-identifies as an underrepresented ethnic minority in their field of research or within their geographical location. One or more of the authors of this paper self-identifies as living with a disability.

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STAR★METHODS

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to the lead contact, Ran Nir-Paz (ran.nir-paz@mail.huji.ac.il).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- Patient information summary is provided in the supplemental data file (see [Data S1](#)).
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Patient referrals

Sixteen patients with a variety of refractory *P. aeruginosa* infections were treated with PASA16 phages between 2019 and 2022 in Israel and abroad. Patients from Israel were referred by their physicians to the Israeli Phage Therapy Center (IPTC), Prof. Ran Nir-Paz at Hadassah-Hebrew University Medical Center, and requests from abroad came either to IPTC or to Adaptive Phage Therapeutics Inc. (APT). Phage therapy was administered alongside standard-of-care antibiotics and surgical interventions as necessary. In 14 cases, bacterial cultures from sites of infection were used for phage testing at the IPTC or Adaptive Phage Therapeutics (APT).

Two additional patients, (Cases 15 and 16) with *P. aeruginosa* infections were enrolled into a clinical trial in Australia (Standardised treatment and monitoring protocol to assess safety and tolerability of bacteriophage therapy for adult and pediatric patients (STAMP study), following the process outlined in the protocol for participant recruitment and eligibility screening.²⁰ Bacterial cultures from sites of infection were sent to Iredell Lab at Westmead Institute for Medical Research for phage testing. Phage production, purification and end-point quality control (QC) was performed at WIMR.

Cases and data collection

In order to collect data on patients treated with PASA16 we reached out directly or through the APT clinical team to the physicians who had provided care for them. Requests for both narrative summaries and clinical research files (CRF) were sent. All but one of the physicians responded with detailed case summaries. The one case for which we had no additional information was excluded from this study (Case 12). Information on gender and age status was reported by the treating physician. ([supplemental information: Table S2 and Data S1 file](#)).

Ethics approval

Approvals for treatments varied according the regulations among the different countries. In Israel, both IRB/EC (Local Helsinki committee) and Ministry of Health approval were required. The publication of the study was carried out following the IRB approvals at Hadassah, with the respective approval numbers being 0420-12-HMO and 0309-19-HMO. In the USA, The following approvals were obtained: UCSD IRB approval number 200163, Johns Hopkins University approval number

IRB00276217, UPMC emergency approval number EA21110067, University of Maryland eIND 19225, RGH IRB approval number 2172. The Food and Drug Administration's (FDA) compassionate use expanded access pathway was used. In Australia, treatment approval was granted by the SCHN drug committee and hospital executive board under the Therapeutic Goods Administration Special Access Scheme. This approval was obtained with patient consent and in accordance with the guidelines set by the Human Research Ethics Committee (HREC)²⁵ or under HREC approval for the STAMP protocol.²⁰ The trial is registered under the number ACTRN12621001526864. All reported patients were asked to provide informed consent prior to the initiation of therapy.

Consent from patients for the publication of photos in case 2 and case 7 has been obtained as well.

METHOD DETAILS

Phage PASA16

In 2018, PASA16 was isolated from a garden soil sample in Jerusalem and purified using the phage titration method, by the Israeli Phage Therapy Center (IPTC). Its genome sequence was deposited in the GenBank (Accession MT933737.1) and stored in the Israeli Phage Bank (IPB).^{31,32} PASA16 received its name after the target bacteria it was derived from, *Pseudomonas aeruginosa*. The acronym also includes the initials of the student who isolated it, Sivan Alkalay, and the corresponding sample number; 16. The genome of PASA16 is 66,127 base pairs (bp) in length, exhibiting a GC content of 55.6%, and it comprises 90 coding DNA sequences.³¹ The phage received the following taxonomy: *Duplodnaviria*, *Heunggongvirae*, *Uroviricota*, *Caudoviricetes*, *Caudovirales*, *Myoviridae*, and *Pbunavirus*. It was initially used to treat a patient with chronic osteomyelitis, then produced in larger quantities by Adaptive Phage Therapeutics Inc. (APT) and made available for compassionate treatments of patients around the globe.

Phage production for intravenous use by Adaptive Phage Therapeutics Inc.

All pharmaceutical grade PASA16 bacteriophage formulations were manufactured by Adaptive Phage Therapeutics (APT) (Gaithersburg, Maryland) using proprietary manufacturing processes. Briefly, the host bacterial strain (PaNP14) was grown in non-animal origin TSB media. Bacterial growth was monitored via spectrophotometric methods until growth reached early to mid-log phases. The bacterial culture was infected with phage stocks and allowed to continue incubation. Upon detection of lysis, the phage lysate was harvested and cleared via filtration. The phage lysate was further purified and formulated via APT's proprietary processes. The formulated phage was aseptically filled into 2 mL single-use vials using a fully enclosed, robotically-operated ISO 5 classified isolator (VanRx Microcell). Vials were visually inspected and a subset was pulled for Quality Control testing. Final drug product vials were frozen at -80°C.

Two batches were used in the studies described. The first batch had a titer of 1.7×10^{11} PFU with 170 endotoxin units/mL. The second batch titer was 4.7×10^{10} PFU with 1.8 endotoxin units/mL. Both batches of PASA16 were provided in Plasma-LYTE A and 20% (v/v) of glycerol (USP grade, as a cryo-preserved) and stored frozen at -80°C. Detailed specifications of each batch are provided in [Table S1](#).

Other phages reported here were: *Pseudomonas* phages PaWRA02Phi83 and PaWRA02Phi11 were sourced from Walter Reed Army Institute of Research; The

Klebsiella phage KpKH52Phi07B and Staphylococcus phage SaWlQ0488Phi1 were sourced from the Naval Medical Research Center's (NMRC) Biological Defense and Research Directorate (BDRD). All phages were manufactured and provided to the treating institutions by APT.

Phage production for topical use by Iredell Lab at Westmead Institute for medical research

PASA16, Pae7 and Pae8 formulations used in Cases 15 and 16 were manufactured by Iredell team at Westmead Institute for Medical Research (WIMR, Westmead, Australia). Both Pae7 and Pae8 are Pseudomonas phages isolated locally at Westmead Institute for Medical Research by members of the Iredell Lab. These phages have not been deposited in GenBank yet. Briefly, the host bacterial strain (JIP697) was grown in Luria Bertani Broth (Lennox). JIP697 at OD 0.2 was added to 150 mL of LB (Lennox) and incubated at 37°C for 20 minutes, 225 rpm. Following incubation, the bacterial culture was infected with phage stock at multiplicity of infection (MOI) 1 and incubated at 37°C for 16 hours, 225 rpm. The phage lysate was clarified via centrifugation and filtration without using Cesium Chloride (CsCl) purification during centrifugation. The batch of PASA16 clarified lysate used in Case 15 and Case 16 had a titer of 4.0×10^9 PFU/mL and was diluted to the dose concentration of 1.0×10^9 PFU/mL. The pH of the diluted clarified lysate was 7.14. The diluted clarified lysate was aseptically filled into 2 mL single-use vials, and 1 vial was subjected to sterility testing. Endotoxin testing was not performed for the topical formulation used in Cases 15 and 16.

Phage susceptibility analysis

Phage susceptibility testing (termed also Clinical Phage Microbiology, CPM) was performed at both IPTC and APT using different protocols. At the IPTC (Table 1), three sorts of assays that have been described previously³³ were used: spot assays, plaque-forming units (PFUs), and kinetic assays in combination with antibiotics to test for possible antagonistic effects. At APT (Table 2), phage kinetic assays were performed to measure the ability of PASA16 along with antibiotics, to inhibit bacterial growth. Kinetic assays were performed in planktonic and biofilm cultures when possible with minor modifications of a previously-described Biolog OmniLog® method.^{34,35}

For Case 15 and Case 16 (Table 3), phage testing was performed in the Iredell Lab at WIMR. Currently, phage susceptibility testing and matching consists of spot test assays, efficiency of plating (EOP) and kinetic assays with different phages tested individually and in combination to test for synergistic effects.^{36,37} For each patient prior to treatment, phage susceptibility testing was performed to ensure sensitivity for the phage.^{11,26} Only patients in whom susceptibility testing indicated possible effectiveness of phage against the *P. aeruginosa* isolate were treated.

Antimicrobial resistance definition

In all cases, the CLSI criteria for defining bacterial resistance were used.³⁸ Standardized definitions were employed to assess antimicrobial resistance, as previously described.³⁹ Briefly, Multi-drug resistance (MDR) was defined as the lack of susceptibility to at least one antimicrobial agent in three or more categories. Extensive-drug resistance (XDR) was defined as the lack of susceptibility to at least one antimicrobial agent in all but two or fewer categories. Pan-drug resistance (PDR) was defined as the lack of susceptibility to all antimicrobial agents across all categories.

Outcome definitions

The outcomes of phage treatment were divided into two categories: clinical and microbiological outcomes. Clinical outcomes were assessed by the treating physicians and consisted the following terms.⁴⁰ Recovery which was defined as no clinical signs of infection detected by physical examinations or PET/MRI imaging and negative microbiological cultures from site of infection; Remission defined as no clinical signs of infection or no imaging findings that suggest an active infection is present, however bacterial cultures were not obtained during the entire follow-up period. Failure was defined as no improvement in clinical signs/symptoms or positive microbiological cultures from site(s) of infection. In the cases where no cultures were performed, the microbiological outcome was described as not specified.

Adverse effects and safety were not collected on a specific predefined protocol. In each case, the reporting center included the adverse effects as part of their clinical cases treatment course and description.

QUANTIFICATION AND STATISTICAL ANALYSIS

Simple descriptive statistics was performed using Microsoft Excel. No other additional statistical analysis was performed.